

James Koester

Finance & AI/Machine Learning Professional | Koesjame@gmail.com | [LinkedIn](#) | [GitHub](#) | Saint Louis, Missouri | Cell: 314-223-9439

SUMMARY

Senior Data Science & Machine Learning professional with a background in financial operations. Proven track record of architecting end-to-end ML solutions—from data pipeline engineering and predictive modeling to production-grade GenAI deployment. Expertise in MLOps, LLM integration, and financial risk analytics. Committed to driving competitive advantage at the intersection of AI and insurance/reinsurance.

TECHNICAL SKILLS

AI/ML & NLP: BERT, Transformers | LLMs | Sentiment Analysis | TensorFlow | Keras | Scikit-learn | PyTorch | LightGBM

Language & Tools: Python (Pandas, NumPy) | Jupyter | Excel (MACROs, VLOOKUPS)

Cloud & Infrastructure: AWS (EC2, S3) , Docker, CI/CD, MLOps (MLflow,DVC)

NLP Specialized: spaCy | NLTK | Tokenization | Lemmatization | TF-IDF | Vectorization

Databases: SQL | Oracle Developer | Snowflake (Familiar)

Analytics: Statistical Modeling | Root-Cause Analysis | Power BI (Familiar) | Matplotlib | Seaborn

Techniques: Data Cleaning | Automation | Machine Learning (Deep Learning, CNNs) | Predictive Modeling | Data-Driven Decision Making | Cross-Functional Collaboration

Other: Microsoft Office | BETA Trading System | GitHub

EXPERIENCE

Senior Operations Associate - Cost Basis | Wells Fargo Advisors | Jan 2024 - Present

- Championing model governance and data integrity by applying rigorous validation standards to automated financial reporting systems.
- Ensure accurate 1099-B distribution for brokerage accounts.
- Led QA efforts for offshore teams handling option premium adjustments.
- Innovated workflow improvements that enhanced team accuracy and alignment with internal equity and operational standards.

Operations Associate - Cost Basis | Mar 2022 - Jan 2024

- Leveraged Excel (Pivot Tables, VLOOKUP, Macros) to streamline account updates.
- Audited 1099 tax documents for accuracy and compliance.
- Led team training and created comprehensive SOPs for handling complex corporate actions, ensuring standardized processing and improved operational consistency across the department.

Operations Associate - Trade Support | Stifel Financial | Aug 2021 - Jan 2022

- Covered trade resolutions for an equity and options trading desk (\$50MM daily trading volume) for a broker/dealer. Assisted over 30 Financial Advisors and multiple SMA Managers daily for trade resolution questions, rejected trades, and other complex trading issues.
- Ensured minimal disruption to clients' portfolios and a high standard of customer service.

Customer Service Specialist | Frito-Lay | May 2012 - Aug 2021

- Hit weekly sales goal of \$20k over 95% of the time due to high levels of customer retention and satisfaction for two corporate vendors.

TECH PROJECTS

Olympic Medal Predictor (Capstone) | MLOps & GenAI | [Github](#)

- **MLOps Pipeline:** Engineered an end-to-end MLOps pipeline, achieving 88% model accuracy through automated experiment tracking (MLflow) and rigorous comparative model versioning.
- **LLM Integration:** Integrated an LLM-based conversational interface (Llama-3.3) to parse unstructured user inputs into structured data for inference.
- **Production Deployment:** Containerized the application using Docker, ensuring environment consistency and scalability for real-time inference.
- **Production Resolution:** Resolved critical latency and hallucination issues in production by implementing a RAG-based caching layer and configuring strict system prompts/guardrails. When drift was detected during evaluation, I remediated the pipeline by updating the feature ingestion layer to align with current data distributions, ensuring model reliability and compliance with accuracy standards.

Corporate Action Automation Processes | Wells Fargo Advisors | 2024-Present

- **Wash Sales & Reclass Automation:** Independently developed SQL scripts to identify Wash Sales Out of Balance (OOB) and Reclass Exceptions. Identified and resolved discrepancies in cost-basis reporting, ensuring 100% compliance with IRS 1099-B regulations.
- **Reporting Efficiency:** Built automated SQL workflows that bypassed standard morning reports, reducing manual adjustments by 30%.

Sentiment Analysis with BERT & Transformers | TripleTen | 04/2026 [Colab](#)

- **Contextual Modeling:** Developed a multi-stage NLP pipeline to classify sentiment in IMDB reviews, progressing from TF-IDF baselines to BERT (Bidirectional Encoder Representations from Transformers).
- **Optimization:** Integrated spaCy for advanced lemmatization and utilized GPU acceleration (T4) on Google Colab to extract high-dimensional embeddings.
- **Performance:** Achieved an ROC AUC of 0.89, demonstrating superior contextual understanding of complex, mixed-sentiment text compared to traditional frequency-based models.

Deep Learning for Insurance Risk Analytics | TripleTen | 01/2026 [Github](#)

- **Predictive Modeling:** Developed hidden layers of Neural Networks (64→32→1) using TensorFlow/Keras to predict medical insurance charges based on demographic and lifestyle risk factors.
- **Model Optimization:** Implemented EarlyStopping and reduced the Learning Rate, achieving a Mean Absolute Error (MAE) of \$2,915.18 while ensuring model generalization and preventing overfitting.

CERTIFICATIONS

AI & Machine Learning Certificate - TripleTen (Issued June 2026)

Securities Industry Essentials (SIE) - FINRA (Issued May 2025)

EDUCATION

Webster University | St. Louis, MO | Master of Science: Finance | MBA

Dominican University | River Forest, IL | Bachelor of Science: Mathematics | Minor: Business